

Ayush Kashyap

Chandigarh, India

ayushkashyap0211@gmail.com — +91 9163519554 — github.com/AyushKashyapII

Education

Punjab Engineering College, Chandigarh
B.Tech in Mechanical Engineering

2023 – 2027
CGPA: 8.28

Technical Skills

Languages: JavaScript, TypeScript, Go, Python, C++, C#, Java

Frameworks: React, Next.js, Node.js, Express.js, React Native

AI/ML: TensorFlow, OpenCV, LangChain, LangGraph, Reinforcement Learning

Systems & Tools: Git, Docker, Redis, PostgreSQL, Firebase, Vercel, WebAssembly

Experience

Software Engineer, Constructure AI

Dec 2025 – Apr 2026

- Built and shipped **backend APIs and services** for LLM-driven applications, enabling structured interaction between models, data sources, and external tools.
- Designed and implemented **scalable data pipelines** for processing large 3D construction datasets, including ingestion, transformation, storage, and retrieval.
- Developed **AI orchestration workflows** that coordinate multiple agents and services, improving system reliability and modularity.
- Implemented **hybrid retrieval systems (RAG)** combining semantic search and structured queries, improving retrieval accuracy and response quality.
- Optimized **system performance and latency** through efficient data handling, caching strategies, and asynchronous workflows.

Projects

GoChess Engine

[Demo](#)

- Built a chess engine using **Minimax with Alpha-Beta Pruning**, reducing search space and improving decision efficiency.
- Implemented **Zobrist Hashing** and **Transposition Tables** to cache evaluated states and avoid redundant computation.
- Added **Quiescence Search** to improve evaluation stability in dynamic positions.
- Compiled the engine to **WebAssembly (WASM)** for browser execution and integrated with a web UI.
- Parallelized computation using **Web Workers**, improving move evaluation speed.

PyGit – Git Clone

[GitHub](#)

- Implemented core Git internals including **Blob, Tree, and Commit objects** using content-addressable storage.
- Built **SHA-1 hashing-based storage** and commit DAG traversal for efficient version tracking.
- Designed a CLI supporting core Git commands: **init, add, commit, log**.

Pacman AI (Reinforcement Learning)

[GitHub](#)

- Built a reinforcement learning agent using **Q-Learning / DQN** for autonomous gameplay.
- Designed state representations and reward shaping to improve learning efficiency and convergence.
- Optimized policy learning to improve overall performance and stability.

Positions of Responsibility

Organizing Team, PEC ACM Hackathon

Sept 2025

Managed logistics and coordination for a 24-hour hackathon.